

Applications of Right Triangles

Calculator Approximations

When dealing with applications involving trigonometric functions, contrary to results so far, it's quite rare that we can determine exact values. In these cases, we will use a calculator's numerical capabilities to determine reasonable approximations for the values in question. For example, most calculators can easily evaluate sine, cosine, and tangent. However, when working with secant, cosecant, or cotangent, we must first manipulate these functions into a form the calculator can understand.

EXAMPLE 1

Give an approximate value for each expression. Round your answer to the nearest hundredth.

(a) $\sin(23^\circ 16' 22'')$

$$\sin\left(23 + \frac{16}{60} + \frac{22}{3600}\right)$$

$$\approx 0.395109...$$

$$0.40$$

(b) $\sec(-95.25^\circ)$

$$= \frac{1}{\cos(-95.25)}$$

$$\approx -10.928768...$$

$$-10.93$$

(c) $\cot(45')$

$$= \frac{1}{\tan\left(\frac{45}{60}\right)}$$

$$\approx 76.390009...$$

$$76.39$$

Inverse Trigonometric Functions

The topic of inverse trigonometric functions can be involved, and frankly, not something we need to worry about at this point in our text. However, to avoid falling prey to some of the pitfalls surrounding the inverse trigonometric functions, we'll restrict our focus to right triangle discussions only. That is, we will only consider inverse trig functions as they apply to acute angles for now.

In general, given a function f and its inverse f^{-1} , if $f(a) = b$, then $a = f^{-1}(b)$. We'll informally refer to this as writing the "inverse form" for a given statement. With regard to inverse sine, inverse cosine, and inverse tangent, we'll use the following notation:

$$\sin(x) = y \leftrightarrow \sin^{-1}(y) = x$$

$$\cos(x) = y \leftrightarrow \cos^{-1}(y) = x$$

$$\tan(x) = y \leftrightarrow \tan^{-1}(y) = x$$

One helpful way to think about the above statements is to consider x as an angle

measure and y as a real number representing the function value for that particular angle.

EXAMPLE 2

For each equation below, write the inverse form and find x or y . Assume x is an acute angle, measured in degrees. Give exact answers whenever possible, and round any approximate solutions to the nearest hundredth.

(a) $\sin(x) = \frac{\sqrt{2}}{2}$ ← QI chart!

$$x = 45^\circ$$

$$\sin^{-1}\left(\frac{\sqrt{2}}{2}\right) = 45^\circ$$

(b) $\sec(x) = \frac{7}{4} \rightarrow \cos(x) = \frac{4}{7}$

$$\cos^{-1}\left(\frac{4}{7}\right) = x$$

$$x \approx 55.15009542\dots$$

$$x = 55.15^\circ$$

(c) $\cos^{-1}(y) = 60^\circ$

$$\cos(60^\circ) = y \leftarrow \text{QI chart!}$$

$$y = \frac{1}{2}$$

$$\cos^{-1}\left(\frac{1}{2}\right) = 60^\circ$$

(d) $\tan^{-1}(y) = 32.235^\circ$

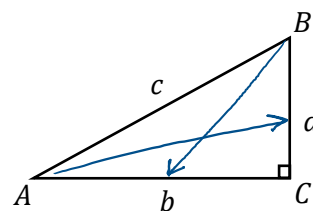
$$\tan(32.235^\circ) = y$$

$$y \approx 0.63058705\dots$$

$$y = 0.63$$

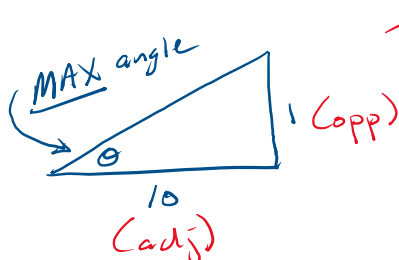
Solving Right Triangles

To solve a triangle means we are to find the measures for all angles and sides. As a reminder, for the triangle ABC , it is customary to label the side opposite angle A as side a , the side opposite angle B as side b , and in the case of right triangles, we'll often denote the right angle as angle C with the hypotenuse as side c .



EXAMPLE 3

- (a) Building codes often regulate the steepness of wheelchair access ramps for public buildings. For example, suppose you are to design a ramp where the building code requires a maximum vertical rise of 1 unit for every 10 units of horizontal distance. What would be the largest possible incline angle for the ramp?



$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{1}{10}$$

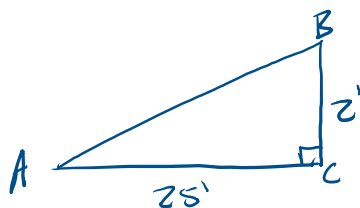
$$\theta = \tan^{-1}\left(\frac{1}{10}\right)$$

$$\theta \approx 5.710593137... \rightarrow \theta = 5.7^\circ$$

- (b) Assume you've been contracted to design and build a wheelchair access ramp. The ramp must start at the sidewalk at a horizontal distance of 25 feet from the door's landing. If the landing is 2 feet above the level of the sidewalk, is there enough space to build a ramp that satisfies building codes?

2' vertical requires a minimum of 20' horizontal ✓

- (c) Designers must supply city engineers with detailed plans to obtain a permit to build this ramp. Sketch a right triangle as your overall design for the ramp and record all angle and side measures for this design.



We know

$$\begin{aligned} a &= 2 \text{ ft} \\ b &= 25 \text{ ft} \\ C &= 90^\circ \end{aligned}$$

We need

$$\begin{aligned} A &= \tan^{-1}\left(\frac{2}{25}\right) \text{ (approx } 4.57^\circ) \\ B &= \tan^{-1}\left(\frac{25}{2}\right) \text{ (approx } 85.43^\circ) \\ c &= \sqrt{629} \text{ ft (approx } 25.08') \end{aligned}$$

Find c (Pythag.)

$$c^2 = (25)^2 + (2)^2$$

$$c = \sqrt{629}$$

$$c \approx 25.07987241...$$

Find A

$$\tan(A) = \frac{2}{25}$$

$$A = \tan^{-1}\left(\frac{2}{25}\right)$$

$$A \approx 4.57392126...$$

*Find B

$$\tan(B) = \frac{25}{2}$$

$$B = \tan^{-1}\left(\frac{25}{2}\right)$$

$$B \approx 85.42607874...$$

Check $A + B + C = 180$

$$4.57 + 85.43 + 90 = 180 \checkmark$$

*used given info to find angle B

Angle of Elevation and Angle of Depression

DEFINITION: Angle of Elevation and Angle of Depression

Any angle measured using the horizontal as the initial side is called an **Angle of Elevation**.

Typically, an angle of elevation is the positive acute angle formed by the horizontal and the line of sight to an object positioned above the horizontal. Any object below the horizontal will produce a negative angle of elevation.

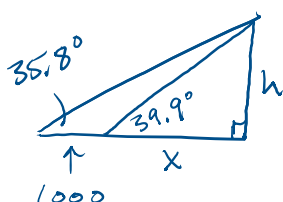
A positive acute angle measured from the horizontal downward is often called an **Angle of Depression**.

It should be noted that an angle of depression is the magnitude (absolute value) of any negative angle of elevation. In this text, we'll restrict the use of the term *angle of elevation* to positive acute angles only.

This terminology is often used in surveying and other related fields. Before the days of GPS and lasers, surveyors could obtain fairly accurate elevation measurements using a simple clinometer.

EXAMPLE 4

A surveyor is charged with estimating the elevation of a particular mountain. He completes the work in a small, relatively flat valley with a known elevation of 4500 ft above sea level. First, using a clinometer, he measures the angle of elevation to the summit to be 35.8° . Then, after walking 1000 feet directly toward the mountain, he takes another measurement. This time the angle of elevation is 39.9° . Using this information, determine the elevation at the peak of the mountain.



$$\tan(35.8) = \frac{h}{1000+x} \rightarrow h = (1000+x)\tan(35.8)$$

$$\tan(39.9) = \frac{h}{x} \rightarrow h = x \tan(39.9)$$

$$x \tan(39.9) = 1000 \tan(35.8) + x \tan(35.8)$$

$$x \tan(39.9) - x \tan(35.8) = 1000 \tan(35.8)$$

$$x (\tan(39.9) - \tan(35.8)) = 1000 \tan(35.8)$$

$$x = \frac{1000 \tan(35.8)}{\tan(39.9) - \tan(35.8)} \approx \frac{721.2227463}{0.114907036}$$

$$x \approx 6276.576017$$

$$* h \approx 6276.576017 \tan(39.9)$$

$$h \approx 5248.03214$$

$$\text{elevation} \approx 4500 + 5248 = \underline{\underline{9748 \text{ ft}}}$$

